

SHIVIN RAJESH ANAND

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EDUCATION

University of Florida

B.S. Computer Engineering

Gainesville, FL

May 2030

SKILLS

Languages & Tools: Python, C++, Java, JavaScript, HTML/CSS, MongoDB, Embedded C, Git, Linux, OpenCV, I2C Protocol

Engineering & Design: Circuit Design, Regression Analysis, Rapid Prototyping, Robotics, Robotic Kinematics, Mechanical Design, Electronics Prototyping, Sensor Integration

Additional: Web Design, Team Leadership, Project Management

EXPERIENCE & INVOLVEMENT

USF

Head Researcher

Tampa, FL

April 2024 – May 2026

- Delivered a validated biomass prediction system with **6.9% MAPE and $R^2 = 0.91$** across **6 independent growth trials** by building an OpenCV computer vision pipeline in Python(HSV conversion, Excess Green Index segmentation, morphological refinement) to autonomously extract canopy area
- Enabled continuous real-time environmental monitoring (CO₂, VOCs, temperature, humidity) by designing and fabricating a custom PCB with I²C sensor architecture, integrating a BME680 sensor with embedded C firmware on a Teensy microcontroller
- Reduced manual harvest requirements to **zero** by developing and iterating a closed-loop autonomous hydroponic growth chamber, combining embedded sensing, computer vision, and predictive regression modeling into a single engineered system

Middleton Robotics

President

Tampa, FL

May 2024 – May 2026

- Ranked in the **top 5% statewide** for offensive power rating by leading mechanical design and rapid prototyping for two FTC robots on Team #3846, iterating drivetrain and manipulator subsystems across build seasons
- Returned the school's FRC program to competition after years of inactivity by rebuilding the mechanical and electrical pipeline, deploying Java on RoboRIO and C++ on VEX controllers with code pushed via Git for wireless robot updates, resulting in **2 national awards and 1 World Championship award** in its first season back
- Qualified two FIRST Tech Challenge teams (1369 and 26588) for state competition by directing robot design reviews and iterative prototyping cycles across mechanical, electrical, and software subsystems
- Expanded competitive participation by **30%** by coordinating **9 teams across 15+ regional competitions** (4 VEX, 4 FTC, 1 FRC), managing engineering resource allocation across divisions

Florida Student Association of Mathematics

Region 4a Coordinator & Webmaster

Tampa, FL

June 2023 – May 2026

- Reduced manual communication overhead by **60%** by developing and maintaining the organization's website in HTML, CSS, and JavaScript, managing code with Git version control and deploying on Linux-based hosting
- Boosted regional engagement by **50%** by creating and managing pathways for **80 Tampa Bay students** into elite competitions (HMMT, PUMaC, ARML, CMIMC) as Regional Coordinator
- Streamlined statewide outreach to **400+ students and advisors** by overseeing communications, event scheduling, and competition logistics infrastructure across Florida

PERSONAL PROJECTS

Inventory & Storefront Web App for Small Business - Independent Project

- Managed real-time inventory tracking for approximately **1,000 products** by independently designing and coding a full-stack web application using HTML, CSS, and JavaScript with a self-built MongoDB database
- Enabled instant stock updates during high-traffic exhibitions by implementing a QR-code scanning system linked to each product tag, allowing real-time add/subtract inventory actions
- Separated admin and customer experiences by building two distinct authenticated views: an admin-side interface for inventory management and a public-facing storefront displaying live product availability sourced from Instagram content

AWARDS & HONORS

- NASA Kennedy Space Center Award:** Southeastern Science and Engineering Fair (SSEF); recognized for research demonstrating the spaceflight viability of the USF hydroponic growth chamber project 2026

LANGUAGES

Hindi | Gujarati | Spanish (Basic)

CERTIFICATIONS

Technical: Autodesk Inventor, AutoCAD & Fusion 360 (Certified User) | Linux Essentials (LE-1) | Microsoft Certified: Azure Fundamentals | CompTIA ITF+

Engineering & PM: RECF Robotics, Programming, Electrical, Mechanical, Fundamentals of Engineering | PMI Project Management Ready

Business/Communication: Master Certification in Entrepreneurship | ESB: Entrepreneurship and Small Business v.2 | Pearson Critical Career Skills: Professional Communication